SPSC: Zero-Shot Cross-Subject EEG-to-Image Reconstruction via Semantic Preservation and Subspace Calibration

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Abstract

Pretrained vision-language models and diffusion generators have driven rapid progress in brain-to-image generation, while EEG-based reconstruction is still constrained by low signal-to-noise ratio and cross-subject semantic shifts. Zero-shot cross-subject reconstruction poses even greater challenges: EEG semantic embeddings learned from source subjects deviate from the manifold of image embeddings of the unseen target subject, particularly when neither target labels nor target image embeddings are available. To address this issue, we introduce the Semantic Preservation and Subspace Calibration (SPSC) framework, which first uses Object Mask Contrastive Regularization (OMCR) with object identity for backend training, and then constructs Subspace Candidate Gate (SCG) to achieve label-free semantic correction for target subjects during inference. Specifically, OMCR regularizes the conditioning space with object-level positive samples to preserve object-level feature neighborhoods within the reconstruction prior and low-level branch, while SCG exploits training image prototypes and batch statistics of unlabeled target EEG to select candidates for each task within subspaces of image prototypes. Under the leave-one-subject-out (LOSO) setting across ten subjects on the THINGS-EEG dataset, SPSC improves the cross-subject reconstruction from 0.552 to 0.713 on CLIP and from 0.612 to 0.745 on Alex.5. These significant gains indicate that the semantically calibrated conditions can better support generation for the target subjects under label-free inference.

Introduction

Visual neural decoding aims to recover perceived image content from brain activity. Early work established the feasibility of image reconstruction from fMRI because its spatial specificity provides stable visual responses (Naselaris et al. 2009). Recent systems further connect neural activity with pretrained vision-language representations and diffusion generators, enabling reconstruction of natural images from fMRI and MEG (Takagi and Nishimoto 2023; Ozcelik and VanRullen 2023; Chen et al. 2023; Scotti et al. 2023; Benchetrit, Banville, and King 2024). However, these results show that a powerful generator is not sufficient by itself: the neural signal must be converted into semantic conditions that remain meaningful to the generator.

EEG is well-suited for this task because it is portable, inexpensive, and temporally precise. Large-scale datasets such as THINGS-EEG provide dense responses to diverse natural images and make EEG-based visual decoding measurable at the object concept level (Hebart et al. 2019; Grootswagers et al. 2022; Gifford et al. 2022). Recent EEG methods align EEG signals with pretrained visual embeddings and use the aligned representation for retrieval, generation, or both tasks (Song et al. 2023; Bai et al. 2023; Singh et al. 2023; Li et al. 2024; Wu et al. 2025; Zhang et al. 2026; Dai et al. 2026). These studies establish pretrained visual spaces as an effective interface for EEG decoding, but this interface remains vulnerable when applied to unseen subjects without labeled calibration images.

However, cross-subject reconstruction faces two core challenges. First, anatomical, physiological, and recording differences induce subject-dependent EEG shifts, thus a source-trained projector may yield target embeddings whose global distribution and local coordinates shift away from the image embedding manifold (Kobler et al. 2022; Huang et al. 2025). Second, existing EEG–image objectives mainly optimize one-to-one trial–image alignment (Song et al. 2023; Li et al. 2024; Zhang et al. 2026), without explicitly preserving relationships among images sharing the same object concept. Insights from representational geometry offer a feasible solution: categorical and exemplar-level structure can be recovered from single-trial EEG recordings (Kaneshiro et al. 2015), while fMRI studies reveal that visual cortical responses can be aligned into a shared representational space across different individuals, even given idiosyncratic response topographies (Haxby et al. 2011). These findings motivate us to preserve object-level neighborhoods throughout training and calibrate subject embeddings within a shared visual semantic space at inference.

Therefore, we propose the Semantic Preservation and Subspace Calibration (SPSC) framework for zero-shot cross-subject EEG-to-image reconstruction, which introduces two core modules: Object Mask Contrastive Regularization (OMCR) and Subspace Candidate Gate (SCG). OMCR regularizes this conditioning space with object-level positive samples, thus it maintains pairwise EEG-image alignment and ensures EEG responses corresponding to the same object concept remain semantically adjacent. At inference time, SCG calibrates the target subject's EEG semantic embeddings after the projector. It builds candidate semantic conditions within PCA subspaces of image prototypes using

training image prototypes and batch statistics of unlabeled target EEG data. A label-free confidence gate then selects a valid candidate for retrieval or reconstruction.

Empirically, SPSC improves cross-subject reconstruction performance under a leave-one-subject-out (LOSO) protocol with ten subjects on the THINGS-EEG dataset: raising CLIP from 0.552 to 0.713 and Alex.5 from 0.612 to 0.745. Detailed analyses on LOSO Sub-08 show that low-rank candidates in the PCA subspace of image prototypes yield smoother and more reliable generative conditions, while higher-rank candidates retain fine-grained discriminative structures for the retrieval task.

Our contributions are the following four aspects:

- We introduce a novel zero-shot cross-subject EEG-to-image reconstruction framework that significantly boosts reconstruction performance by calibrating semantic conditions for the target subject without target labels or target image embeddings.
- We propose an Object Mask Contrastive Regularization (OMCR) that preserves semantic neighborhood relationships beyond simple pairwise EEG–image alignment via regularizing conditioning space with object-level positive samples.
- We design the Subspace Candidate Gate (SCG), a label-free candidate gate operating within image subspaces, which leverages training image prototypes and unlabeled target EEG batch statistics to mitigate the subject shift.
- We conduct comprehensive evaluations on the THINGS-EEG dataset, including reconstruction and retrieval tasks. The results validate that the proposed framework outperforms the SOTA methods, and subspace analyses demonstrate that low-rank candidates in the PCA subspace of image prototypes deliver superior conditioning for reconstruction, whereas mid/high-rank candidates better preserve discriminative structures for retrieval.

Related Work

Brain-to-image reconstruction. Modern brain-to-image systems commonly use pretrained visual priors. A neural encoder predicts visual representations, which are then decoded into images by diffusion or unCLIP style generators (Takagi and Nishimoto 2023; Ozcelik and VanRullen 2023; Scotti et al. 2023; Benchetrit, Banville, and King 2024; Ramesh et al. 2022; Rombach et al. 2022). This architecture decouples neural encoding and image synthesis, enabling the generator to inherit large-scale pretrained semantic and visual knowledge. However, cross-subject EEG-to-image reconstruction remains largely underexplored, especially in scenarios without labeled calibration data from the unseen target subject. To this end, we propose a zero-shot cross-subject EEG-to-image reconstruction method via semantic preservation and subspace calibration.

EEG visual decoding. Existing EEG-to-image reconstruction and retrieval methods such as NICE, DreamDiffusion, EEG2IMAGE, ATM, UBP, NeuroBridge, and AVDE align EEG responses with pretrained visual representations (Song et al. 2023; Bai et al. 2023; Singh et al. 2023; Li et al. 2024; Wu et al. 2025; Zhang et al. 2026; Dai et al. 2026). Although these methods enable EEG-based visual reconstruction, most training objectives emphasize paired alignment between single EEG trials and individual image embeddings. However, for datasets containing multiple images per object concept, training solely on isolated pairs restricts the conditioning space to discrete matched pairs, failing to form coherent object-level semantic structures. Accordingly, we introduce OMCR to address this issue by constructing intra-batch positive samples based on shared object identities.

Cross-subject adaptation. Cross-subject transfer has long been a fundamental challenge in neural decoding. For fMRI and related modalities, subject alignment techniques reduce data requirements and improve cross-subject generalization (Scotti et al. 2024; Wang et al. 2024; Gong et al. 2025). However, for EEG, variations stemming from anatomical differences, electrode placement, and trial noise make cross-subject transfer more fragile; NEED also highlights cross-task generalization for EEG-based visual reconstruction (Huang et al. 2025). The proposed SCG further complements these efforts by operating calibration after the EEG projector: it adjusts EEG embeddings toward the manifold spanned by training image prototypes, relying solely on batch statistics extracted from unlabeled target EEG data.

Method

Figure 1 organizes SPSC around two core problems of zero-shot cross-subject reconstruction. Firstly, during generation backend training, OMCR preserves object-level neighborhoods so that the generator receives semantic conditions with stable object structure. Secondly, during target inference, SCG calibrates post-projector EEG semantic vectors from unseen subjects using only training image prototypes and unlabeled target batch statistics. Since retrieval and reconstruction prioritize different semantic scales, SCG exposes two readouts from the same calibrated candidate pool: SCG-Ret for discriminative retrieval and SCG-Recon for stable semantic conditioning in reconstruction. These two modules act at different stages of the pipeline: OMCR regularizes the reconstruction backend, whereas SCG only edits the semantic condition fed into the diffusion prior and generator.

Problem Setup

The reconstruction pipeline first maps EEG to a semantic vector $x \in \mathbb{R}^d$, and then predicts the CLIP image embedding $\hat{h}$ and low-level latent $\hat{z}$ for image generation. This design facilitates the disentanglement between high-level semantic conditioning and low-level image structure.

In cross-subject reconstruction, one subject is held out as the unseen target test subject. During inference, only unlabeled batch statistics from the target subject's EEG semantic vectors are available, while SCG branch adaptation and hyperparameter tuning rely solely on source subjects. Thus, this corresponds to an unlabeled test-batch adaptation setting: target EEG samples are observed in batches, while their stimulus identities and image embeddings are only used for evaluation after inference.

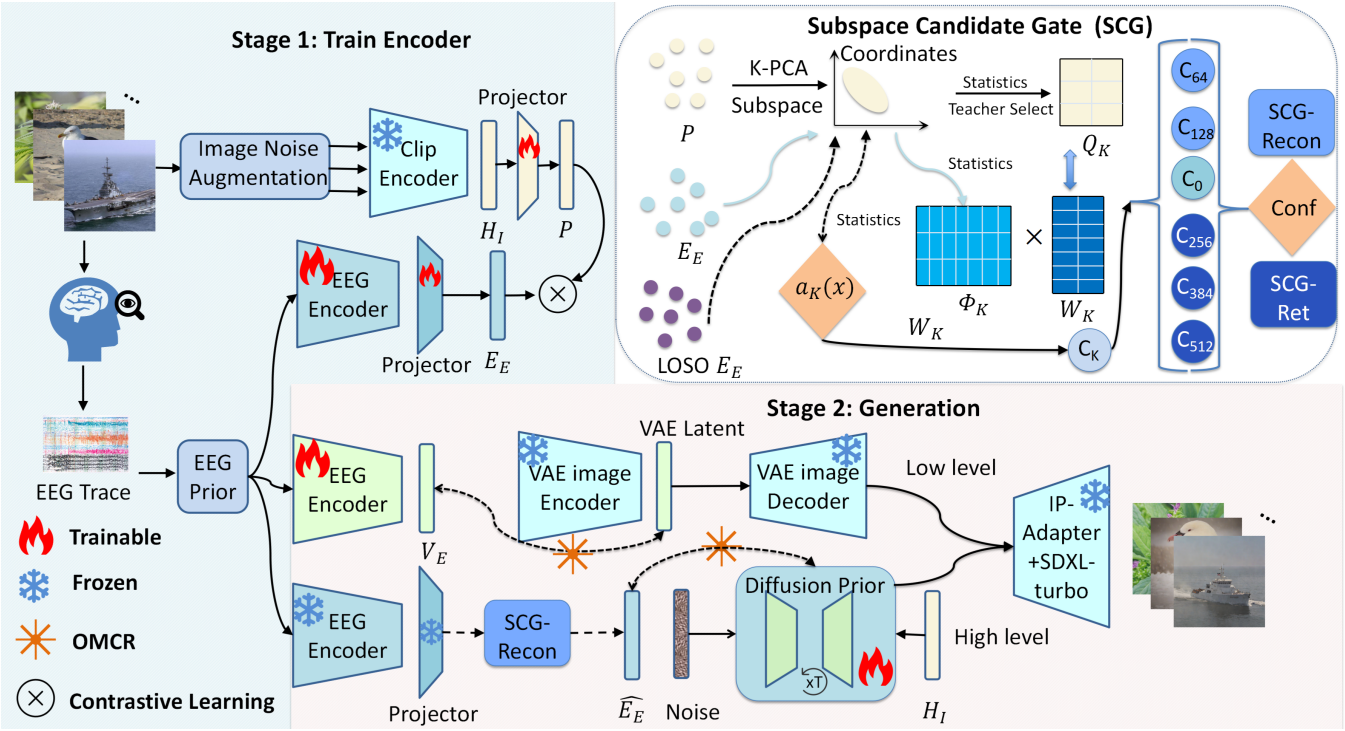


Figure 1: SPSC framework for zero-shot cross-subject EEG-to-image reconstruction. In the stage 1, we train the EEG encoder with the linear projector via constrastive learning and transform EEG to semantic vectors E_E ($x_{i,s}$). Subsequently, SCG uses training image prototypes P and LOSO E_E statistics extracted from the target subject to construct semantic candidates C_K in PCA subspaces of image prototypes, then selects semantic vectors in SCG-Recon or SCG-Ret through a label-free confidence gate. In the stage 2, we adopt SCG-Recon output $\widehat{E_E}$ for image generation, while OMCR preserves object-level neighborhoods within CLIP embeddings H_I of the diffusion prior and the low-level VAE latent features V_E.

Subspace Candidate Gate (SCG)

Aiming to calibrate target-subject semantics after the EEG projector, SCG first learns the optimal transformation that aligns EEG semantic coordinates with the image prototype manifold using data from source subjects. At target inference, SCG instantiates the same correction paradigm using unlabeled target EEG batch statistics and selects one complete semantic candidate for each individual sample. Note that SCG only edits semantic conditions output by the frozen projector; SCG-Ret and SCG-Recon serve as two task-aware readouts derived from the same candidate family: SCG-Ret exposes mid/high-rank candidates for retrieval, whereas SCG-Recon leverages low-rank candidates for reconstruction.

Let N_{tr} be the total number of training stimuli. Let $x_{i,s} \in \mathbb{R}^D$ be the EEG semantic vector output by the projector for the i-th stimulus from the s-th subject, where $D = 1024$. Each stimulus is associated with an image prototype p_i, which refers to the projected semantic embedding of its corresponding training image.

$$P = \{p_i\}_{i=1}^{N_{\text{tr}}}. \tag{1}$$

The prototype bank P provides the image reference used by SCG to compute PCA bases, distribution statistics, branch fitting, and candidate scoring.

Shared subspace and Correlation Alignment (CORAL) variants. The shared subspace determines the image-side coordinates that SCG is allowed to modify. Accordingly, the PCA basis $B_K \in \mathbb{R}^{D\times K}$ is fitted only on the training prototype bank P, where $K \in \{64, 128, 256, 384, 512\}$. For any semantic vector x, we define its coordinate in the PCA subspace of image prototypes and corresponding decoding mapping as

$$u_K(x) = B_K^\top(x - \mu_P), \qquad d_K(u) = \mu_P + B_K u. \tag{2}$$

The prototype coordinate batch $V^K = \{u_K(p_i)\}_{i=1}^{N_{\text{tr}}}$ provides the image-side statistics used to calibrate EEG coordinates within the rank-K subspace. Given a reference coordinate batch R and a target coordinate batch V, we denote μ_R, Σ_R and μ_V, Σ_V as their respective mean and covariance. At each rank, the regularized CORAL operator aligns EEG coordinate statistics to image prototype statistics by whitening z with the reference covariance and recoloring it with the target covariance (Sun, Feng, and Saenko 2016):

$$\begin{aligned}\text{CORAL}_r(z; R, V) = (z - \mu_R)(\Sigma_R + rI)^{-\frac{1}{2}} \\ \cdot (\Sigma_V + rI)^{\frac{1}{2}} + \mu_V.\end{aligned} \tag{3}$$

SCG uses an interpolated CORAL variant

$$T_{r,\rho}(z; R, V) = z + \rho\,(\text{CORAL}_r(z; R, V) - z)\,, \tag{4}$$

where r controls covariance regularization and ρ constrains the strength of alignment toward statistics of image prototype coordinates.

Subspace branch fitting. To learn a reliable semantic correction rule, we exploit the paired EEG-image evidence from source subjects. For this purpose, SCG fits a ridge mapper for each rank across all source subjects to shift EEG coordinates toward image prototype coordinates. Then, to quantify how source EEG coordinates move toward the prototype coordinate batch V^K, we utilize the EEG coordinate batch $U_s^K = \{u_K(x_{i,s})\}_{i=1}^{N_{\text{tr}}}$ from source subject s, which provides the subject statistics. For each rank K, SCG builds CORAL teacher variants under image prototype statistics:

$$t_{i,s}^{K,r,\rho} = T_{r,\rho}(u_K(x_{i,s}); U_s^K, V^K). \tag{5}$$

Among all variants, SCG retains the variant whose decoded vector obtains the highest cosine similarity with the training prototype bank. The ridge input ϕ_K contains a subject-normalized sample coordinate, the subject-specific mean and log standard deviation within the same subspace, along with a constant bias term. This supplies the mapper both the per-sample coordinates and a concise representation of the subject-level distribution shift. Stacking ϕ_K over all source samples yields the matrix Φ_K, and stacking the selected teacher coordinates generates Q_K. The parameters W_K of this mapping branch are fitted by ridge regression:

$$W_K = \arg\min_W \|\Phi_K W - Q_K\|_F^2 \; + \; \lambda \|W\|_F^2 . \tag{6}$$

In implementation, we adopt the closed-form ridge solution with $\lambda = 1$. The fitted W_K will project subject-normalized EEG subspace features into K-dimensional coordinates aligned with image prototypes.

Candidate construction for the target subject. Given an unlabeled EEG batch X, the pipeline for constructing target candidate construction is divided into three distinct components. The first candidate c_0 corresponds to a full-dimensional diagonal alignment (FullDiag), which eliminates global mismatch in mean and feature scale:

$$c_0(x) = \mu_P + \frac{\sigma_P}{\sigma_X + \epsilon} \odot (x - \mu_X), \tag{7}$$

where μ_P, σ_P are prototype statistics and μ_X, σ_X are statistics of the target EEG test batch. For each rank K, we define $U_X^K = \{u_K(x) : x \in X\}$. The CORAL anchor will shift the features of the target batch to a region close to the image prototype distribution within a selected subspace:

$$a_K^{r,\rho}(x) = d_K\left(T_{r,\rho}(u_K(x); U_X^K, V^K)\right). \tag{8}$$

Here, the anchor variant achieving the highest cosine similarity with the training prototype bank is denoted as $a_K(x)$. The target batch supplies the same mean and scale features used by the ridge mapper, producing an image-aligned coordinate prediction $\hat{u}_K(x) = \phi_K(u_K(x))W_K$. The final rank-$K$ candidate $c_K(x)$ applies a controlled residual $\delta_K(x)$ around the selected anchor $a_K(x)$, which are defined as follows:

$$\delta_K(x) = \hat{u}_K(x) - B_K^\top(a_K(x) - \mu_P), \tag{9}$$

$$c_K(x) = a_K(x) + \beta B_K \delta_K(x). \tag{10}$$

The anchor handles distribution-level shift, while the residual carries the source-learned correction; the small blend β preserves the target sample geometry within the selected semantic scale.

Label-free gate. At selection time, SCG performs sample-wise routing over complete candidate vectors via prototype bank confidence:

$$C(x) = \{c_0(x), c_{64}(x), c_{128}(x), c_{256}(x), c_{384}(x), c_{512}(x)\}. \tag{11}$$

Each $c_K(x) \in \mathbb{R}^D$ is a complete semantic condition produced by branch K. Let $s_{(1)}^K \geq \cdots \geq s_{(10)}^K$ be the top 10 cosine similarities between $c_K(x)$ and prototypes in P. The confidence score is

$$\text{Conf}_K(x) = s_{(1)}^K + 2\left(s_{(1)}^K - s_{(10)}^K\right), \tag{12}$$

where, the nearest prototype term $s_{(1)}^K$ measures manifold proximity, and the 2nd term tail gap measures local sharpness. The selector exports the eligible candidate $\hat{c}(x)$ with the highest confidence:

$$K^*(x) = \arg\max_{K \in A} \text{Conf}_K(x), \hat{c}(x) = c_{K^*(x)}(x). \tag{13}$$

Subsequently, we set $A = \{0, 256, 384, 512\}$ to define SCG-Ret, and set $A = \{0, 64, 128\}$ to define SCG-Recon. Both readouts are adopted to evaluate the same candidate pool: the retrieval readout retains higher-rank candidates for fine semantic discrimination, whereas the reconstruction readout restricts selection to lower-rank candidates that provide smoother diffusion conditioning.

Object Mask Contrastive Regularization (OMCR)

OMCR focuses on addressing the existing training-side limitation of pairwise EEG–image alignment: different images from the same object concept can be treated as unrelated negatives. Given predicted features $\hat{r}_i$, target features r_i, and object labels y_i in a batch B, it uses object labels to preserve object-level neighborhoods while retaining the paired image target. The paired cosine anchor loss $\mathcal{L}_{\cos}$ is defined as:

$$\mathcal{L}_{\cos}(\hat{r}, r) = \frac{1}{B}\sum_{i=1}^{B}\left(1 - \cos(\hat{r}_i, r_i)\right). \tag{14}$$

Object labels build the mask $M_{ij} = \mathbf{1}[y_i = y_j]$ in training batches. The final loss $\mathcal{L}_{\text{om}}$ is written as:

$$\mathcal{L}_{\text{om}}(\hat{r}, r, y) = \frac{-1}{2B}\sum_{i=1}^{B}\left[\log\frac{\sum_{j=1}^{B} M_{ij}\exp(s(\hat{r}_i, r_j)/\tau)}{\sum_{k=1}^{B}\exp(s(\hat{r}_i, r_k)/\tau)} + \log\frac{\sum_{j=1}^{B} M_{ij}\exp(s(r_i, \hat{r}_j)/\tau)}{\sum_{k=1}^{B}\exp(s(r_i, \hat{r}_k)/\tau)}\right], \tag{15}$$

where, the s is $cos(\hat{r}, r)$, and the τ is the temperature.

Diffusion prior. Let h_i be the target CLIP image embedding. The diffusion prior follows the standard denoising objective: it receives a noisy embedding δ_t, predicts $\hat{\epsilon}_\theta(\delta_t, t, c_i)$

Method	Metric (%)	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10	Mean
NICE	Top-1	7.60	5.90	6.00	6.30	4.40	5.60	5.60	6.30	5.70	8.40	6.20
(Song et al. 2023)	Top-5	22.8	20.5	22.3	20.7	18.3	22.2	19.7	22.0	17.6	28.3	21.4
ATM	Top-1	10.5	7.10	11.9	14.7	7.00	11.1	<u>16.1</u>	15.0	4.90	20.5	11.9
(Li et al. 2024)	Top-5	26.8	24.8	33.8	39.4	23.9	35.8	<u>43.5</u>	<u>40.3</u>	22.7	46.5	33.8
AVDE	Top-1	<u>14.1</u>	17.0	9.10	15.2	12.5	<u>17.3</u>	7.40	<u>18.5</u>	<u>13.2</u>	18.0	14.3
(Dai et al. 2026)	Top-5	32.2	38.4	21.8	32.5	32.4	<u>38.6</u>	20.4	40.1	<u>33.6</u>	39.3	32.9
NeuroBridge (ViT-H-14)	Top-1	14.0	<u>25.5</u>	<u>14.5</u>	<u>15.5</u>	<u>14.5</u>	13.5	15.5	15.5	12.0	<u>23.5</u>	<u>16.4</u>
(Zhang et al. 2026)	Top-5	<u>39.0</u>	<u>50.5</u>	<u>33.0</u>	<u>39.5</u>	<u>36.5</u>	37.0	40.0	36.0	30.5	<u>52.5</u>	<u>39.5</u>
SCG-Ret (Ours)	Top-1	**24.5**	**28.0**	**20.5**	**18.0**	**19.0**	**27.0**	**22.0**	**24.0**	**23.0**	**30.5**	**23.7**
	Top-5	**47.5**	**53.5**	**47.5**	**42.0**	**44.0**	**54.5**	**46.5**	**46.0**	**48.0**	**62.5**	**49.2**

Table 1: Cross-subject 200-way retrieval accuracy under ten-subject LOSO. **Best** and <u>second best</u> are highlighted.

under the EEG semantic condition c_i, and yields a clean image-embedding estimate $\hat{h}_i$. The loss $\mathcal{L}_{\text{noise}}$ is

$$\mathcal{L}_{\text{noise}} = \mathbb{E}\|\hat{\epsilon}_\theta - \epsilon\|_2^2, \tag{16}$$

with the standard gaussian noise ϵ. Finally, the OMCR $\mathcal{L}_{\text{prior}}$ combines the denoising objective $\mathcal{L}_{\text{noise}}$ with object-level contrastive loss $\mathcal{L}_{\text{om}}$ and the paired cosine anchor $\mathcal{L}_{\cos}$:

$$\mathcal{L}_{\text{prior}} = \mathcal{L}_{\text{noise}} + \mathcal{L}_{\cos}(\hat{h}_i, h_i) + \mathcal{L}_{\text{om}}(\hat{h}_i, h_i, y). \tag{17}$$

Low-level branch. Let z_i be the target VAE latent and $\hat{z}_i$ the predicted latent. The $\mathcal{L}_{\text{low}}$ uses the same object-mask term as an auxiliary regularizer on top of latent regression:

$$\begin{aligned}\mathcal{L}_{\text{low}} =& \|\hat{z} - z\|_1 + \lambda_{\text{mse}}\|\hat{z} - z\|_2^2 \\ &+ \mathcal{L}_{\cos}(\hat{z}, z) + \lambda_o^\ell \mathcal{L}_{\text{om}}(\hat{z}, z, y),\end{aligned} \tag{18}$$

where, λ_{mse} and λ_o^ℓ are weights for balancing scale differences. Because VAE latents also encode layout and texture of individual images, the term $\mathcal{L}_{\text{low}}$ serves only as an auxiliary constraint rather than the main OMCR source of object-level preservation.

Experiments

Datasets and Metrics

Evaluation uses THINGS-EEG, which provides dense RSVP EEG responses to 1,854 visual object concepts with a standard split of 1,654 training object concepts and 200 test object concepts (Gifford et al. 2022). Note that THINGS-EEG is the only publicly available dataset for this work. For reconstruction, we follow prior brain-to-image works (Scotti et al. 2023; Li et al. 2024) and report Alex.2, Alex.5, CLIP, and SwAV, with PixCorr (Pix.) and SSIM included in complete reconstruction tables (Krizhevsky, Sutskever, and Hinton 2012; Caron et al. 2020; Wang et al. 2004). Detailed metric definitions and sanity checks are provided in the Appendix A.

Experimental Settings

EEG and image preprocessing follow the augmentation and semantic-projection protocol of related EEG-to-image work (Zhang et al. 2026). Meanwhile, consistent with prior reconstruction works (Li et al. 2024; Dai et al. 2026), subject 08 is chosen as the primary subject for cross-subject comparison.

Method	Pix. ↑	SSIM ↑	Alex.2 ↑	Alex.5 ↑	CLIP ↑	SwAV ↓
Cross-subject						
Li et al.	0.079	0.374	0.640	0.649	0.556	0.638
SCG-Recon	**0.174**	**0.389**	**0.684**	**0.726**	**0.723**	**0.618**
Intra-subject						
Li et al.	0.160	0.345	0.776	0.866	0.786	0.582
CogCapturer	0.175	0.366	0.760	0.610	0.744	0.577
AVDE	<u>0.188</u>	<u>0.396</u>	<u>0.817</u>	<u>0.889</u>	<u>0.795</u>	<u>0.557</u>
SCG-Recon	**0.304**	**0.397**	**0.868**	**0.905**	**0.888**	**0.538**

Table 2: Subject-08 reconstruction metrics in cross- and intra-subject settings. CogCapturer abbreviates Cognition-Capturer (Zhang et al. 2025).

We implement the reconstruction pipeline using SDXL-Turbo with IP-Adapter (Podell et al. 2023; Ye et al. 2023). Its diffusion prior builds on the U-net with the hyperparameter set to $\tau = 0.07$ (Li et al. 2024). The low-level branch adopts latent regression with low-level OMCR under $\lambda_o^\ell = 0.5$ and $\lambda_{\text{mse}} = 0.01$. For SCG, the CORAL teacher and anchor variants use fixed pairs $(r, \rho) \in \{(2, 0.9), (4, 0.9), (6, 1.0), (2, 1.0), (6, 0.9)\}$, covering CORAL regularization strengths ranging from relatively aggressive to mild. The ridge regularization is configured with $\lambda = 1$, while the residual branch blend is $\beta = 0.125$.

All experiments are implemented in PyTorch 2.5.1 and run on a single NVIDIA RTX 4090 GPU with 24 GB. All reported SCG reconstruction results adopt the same OMCR settings. Additional setting details are provided in the Appendix B and C.

Brain Decoding Performance

Table 1 reports cross-subject 200-way retrieval under ten-subject LOSO. SCG-Ret outperforms the the strongest baseline considered, NeuroBridge, by 0.073 in Top-1 accuracy, indicating improved target-subject semantic separability. Table 2 evaluates Sub-08 reconstruction. In the cross-subject setting, SCG-Recon outperforms Li et al. with gains of 0.167

Method	Stat	Alex.2 ↑	Alex.5 ↑	CLIP ↑	SwAV ↓
Li et al.	Mean	0.590	0.612	0.552	0.648
	Std	0.025	0.021	0.032	0.006
SCG-Recon	Mean	**0.706**	**0.745**	**0.713**	**0.618**
	Std	0.028	0.023	0.027	0.010

Table 3: Cross-subject reconstruction under ten-subject LOSO, reported as mean and population standard deviation.

Method	Alex.2 ↑	Alex.5 ↑	CLIP ↑	SwAV ↓
Base	0.508	0.513	0.527	0.718
Base + OMCR	0.676	0.680	0.679	0.631
+FullDiag	0.677	0.696	0.680	0.627
+SCG-Ret	0.678	0.702	0.694	0.622
+SCG-Recon	**0.684**	**0.726**	**0.723**	**0.618**

Table 4: LOSO Sub-08 reconstruction ablation.

on CLIP and 0.077 on Alex.5; in the intra-subject setting, it achieves the best performance across the reported semantic and perceptual metrics. Table 3 further shows that the reconstruction advantage generalizes across subjects, with ten-subject LOSO gains over Li et al. of 0.161 on CLIP and 0.133 on Alex.5. These results support task-specific readout selection in SPSC: retrieval benefits from higher-rank discriminative coordinates, whereas reconstruction benefits from lower-rank semantic conditions that remain more stable for diffusion-based synthesis. Complete per-subject reconstruction and retrieval results for cross- and intra-subject settings are provided in the Appendix E and F.

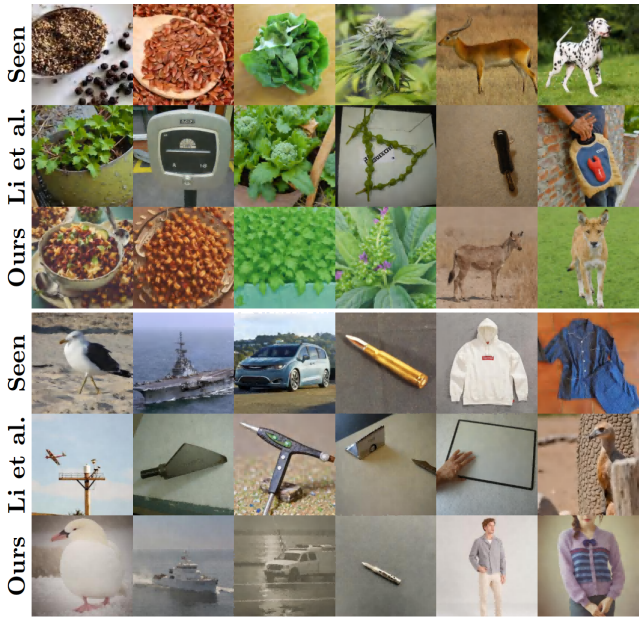


Figure 2: Cross-subject Sub-08 reconstruction examples.

Figure 2 visually illustrates the same trend: calibrated low-rank semantics preserve object identity more consistently than Li et al. under the same generation backend.

Ablation Studies

Reconstruction ablation. Table 4 separates backend preservation from target semantic calibration for cross-subject Sub-08. OMCR delivers the primary performance gain by preserving object-level positives during backend training, increasing the CLIP by 0.152 over Base. When OMCR is enabled, SCG changes only the semantic condition fed into the prior training and inference, and SCG-Recon yields the best reconstruction metrics and provides another 0.044 the CLIP gain. These results validate the intended division of labor between object-level feature preservation and target-subject semantic calibration.

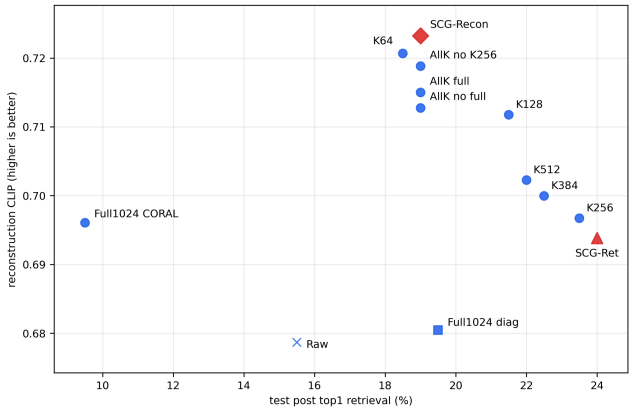


Figure 3: LOSO Sub-08 retrieval–reconstruction trade-off.

SCG candidate behavior. Figure 3 shows that retrieval and reconstruction prefer different semantic scales under the same OMCR backend. K256 and SCG-Ret occupy the retrieval-oriented region, while K64/K128 and SCG-Recon give better reconstruction performance. The two readouts therefore represent task-adapted representations derived from the same candidate semantic pool. On the LOSO Sub-08 test, SCG-Ret reaches a Top-1 score of 24.0, whereas SCG-Recon attains the highest reconstruction CLIP of 0.723. Full candidate-level retrieval and reconstruction metrics are provided in the Appendix D.

Analysis and Discussion

Subspace reliability and subject shift. Figure 4 measures correction strength, semantic SNR, and EEG–image alignment across varying image-PCA ranks. Low ranks yield higher SNR and stronger EEG–image correlations, while higher ranks introduce finer but less stable coordinate representations. This pattern explains why SCG-Recon uses low ranks for generation and SCG-Ret keeps mid/high ranks for retrieval.

Confidence gate and selector behavior. The confidence gate enables label-free routing based on semantic similarity sharpness against training prototypes. The nearest-prototype term measures manifold proximity, while the tail gap captures local sharpness. Table 5 compares this routing rule with

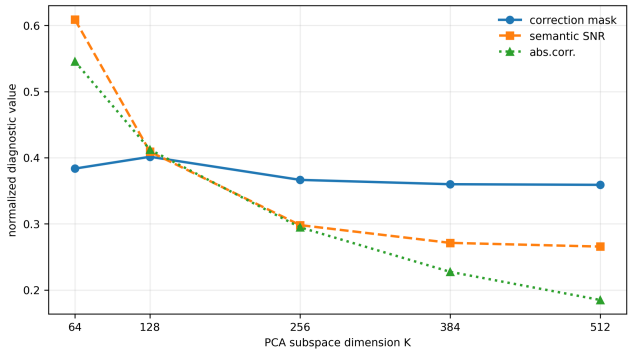


Figure 4: Diagnostics for LOSO training subspace scale. The correction mask estimates the required correction magnitude for each PCA coordinate; semantic SNR is defined as the variance across stimuli divided by the variance across subjects for the same stimulus; abs.corr. denotes the mean absolute Pearson correlation between EEG embedding coordinate means of source subjects and paired image prototype coordinates.

Readout Selector	Top-1 (%) ↑	Top-5 (%) ↑
SCG-Ret Confidence	<u>24.0</u>	46.0
SCG-Ret Fixed k256	23.5	<u>46.5</u>
SCG-Ret Average	21.0	46.0
SCG-Ret Oracle	**27.0**	**47.5**

Table 5: LOSO Sub-08 retrieval selector behavior.

fixed, average, and oracle alternatives; the oracle row serves as a post-hoc diagnostic that evaluates candidate complementarity unavailable to the label-free gate. The gate turns a single SCG candidate family into task-conditioned readouts: SCG-Ret routes toward high-rank semantics for retrieval discrimination, whereas SCG-Recon selects smoother low-rank semantics for generation. Additional selector diagnostics in the Appendix D show the routing behavior for SCG-Recon.

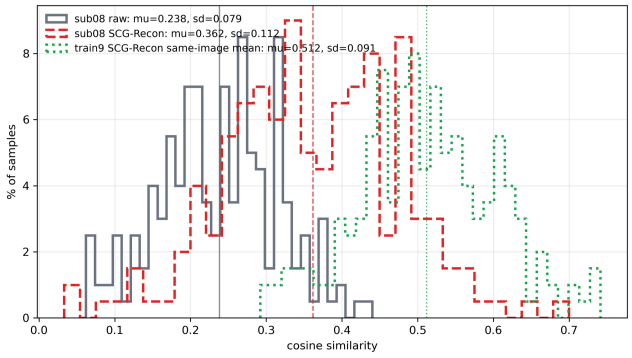


Figure 5: LOSO Sub-08 paired semantic alignment.

Semantic alignment of the image side. Figure 5 relies solely on held-out image embeddings after inference to measure whether the exported semantic condition moves toward the matched test embedding. For each test stimulus, the score corresponds to the cosine similarity between the corrected EEG semantic and the semantic prototype of the test image. Raw Sub-08 semantics generate a mean cosine score of 0.238, while SCG-Recon shifts the target semantics up to 0.362; averaging responses from the 9 training source subjects for the same image reaches 0.512, which acts as an analytical reference for image-specific cross-subject alignment. This comparison with paired images links the reconstruction readout to generator conditioning: it supplies a smoother semantic condition for the target subject that lies closer to the image subspace within the shared embedding space.

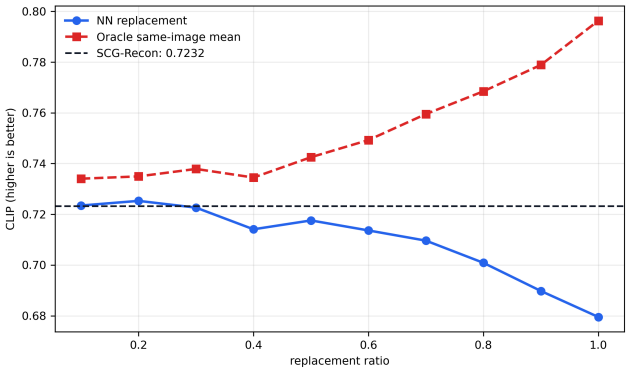


Figure 6: LOSO Sub-08 prior replacement curve.

Cross-subject prior analysis. Figure 6 examines whether source-subject test responses contain reusable image-specific structure. Nearest-neighbor (NN) replacement operates in a label-free way but risks overwriting the target manifold geometry, while oracle replacement uses same-image identity and serves only as an upper-bound reference. NN replacement delivers only marginal gains at low replacement ratios and then underperforms SCG-Recon as more target semantics get overwritten. In contrast, the oracle curve rises with greater access to the test identities. SCG-Recon remains practically deployable since it calibrates the target response itself rather than replacing it with source-subject semantics.

Conclusion

Our SPSC framework addresses zero-shot cross-subject EEG-to-image reconstruction as semantic conditioning for generative models. Specifically, OMCR preserves object-level neighborhoods during generation backend training, and SCG calibrates the target subject's EEG semantics after the projector using training image prototypes and unlabeled target EEG statistics. Under ten-subject LOSO evaluation, the reconstruction readout improves key semantic reconstruction metrics during label-free inference for the target subject. The analyses show that low-rank calibrated semantics better align the target EEG condition with the embedding manifold of the image side for reconstruction, while higher-rank semantics retain discriminative structure for retrieval.

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Appendix

Supplemental Material for "SPSC: Zero-Shot Cross-Subject EEG-to-Image Reconstruction"

A. Metric Definitions and Sanity Check

Following prior brain-to-image evaluation protocols, Alex.2 and Alex.5 denote two-way identification scores computed from AlexNet layers 2 and 5, and CLIP denotes the corresponding identification score computed from CLIP embeddings. SwAV is the feature distance from SwAV-ResNet50, with lower values indicating stronger feature similarity. PixCorr and SSIM quantify low-level image similarity and are reported as auxiliary reconstruction measures.

Condition	PixCorr ↑	SSIM ↑	Alex.2 ↑	Alex.5 ↑	CLIP ↑	SwAV ↓
raw image	0.199	0.284	0.858	0.949	0.995	**0.423**
augmentation	**0.231**	**0.293**	**0.868**	**0.953**	**0.996**	0.443

Table 6: Oracle CLIP-to-SDXL-Turbo sanity check for ViT-H-14 embedding.

The oracle control passes ground-truth ViT-H-14 CLIP embeddings of test images directly to SDXL-Turbo with IP-Adapter, without using EEG. This control confirms that true CLIP image embeddings yield strong semantic scores even when PixCorr and SSIM remain modest, as the generator can preserve object identity while varying texture or layout. Accordingly, PixCorr and SSIM are treated as auxiliary reconstruction measures.

B. SCG Implementation Details

This section gives the implementation details condensed in the main method while preserving the same notation. For compact notation in this appendix, define $u_{i,s}^K = u_K(x_{i,s})$, $v_i^K = u_K(p_i)$, $U_s^K = \{u_{i,s}^K\}_{i=1}^{N_{\mathrm{tr}}}$, and $V^K = \{v_i^K\}_{i=1}^{N_{\mathrm{tr}}}$.

Teacher variant selection. For each source sample, CORAL variants produce teacher coordinates $t_{i,s}^{K,r,\rho} = T_{r,\rho}(u_{i,s}^K; U_s^K, V^K)$. SCG keeps the teacher whose decoded vector is closest to the training prototype bank:

$$(r^*, \rho^*) = \arg\max_{r,\rho} \max_{p \in P} \cos\Big(d_K(t_{i,s}^{K,r,\rho}), p\Big), \qquad \tilde{t}_{i,s}^K = t_{i,s}^{K,r^*,\rho^*}. \tag{19}$$

The selected teacher is anchored to the paired image coordinate by the mix with a weak weight α

$$q_{i,s}^K = (1-\alpha)\tilde{t}_{i,s}^K + \alpha v_i^K, \qquad \alpha = 0.10. \tag{20}$$

Ridge feature construction. The ridge input concatenates the normalized coordinate, source-subject mean, log-scale, and a bias term:

$$\phi_K(u_{i,s}^K) = \left[\frac{u_{i,s}^K - \mu_s^K}{\sigma_s^K}, \mu_s^K, \log \sigma_s^K, 1\right]. \tag{21}$$

Let Φ_K stack these features over source subjects, and let Q_K stack the teacher coordinates $q_{i,s}^K$. The closed-form ridge solution is

$$W_K = (\Phi_K^\top \Phi_K + \lambda I)^{-1} \Phi_K^\top Q_K, \qquad \lambda = 1. \tag{22}$$

With S source subjects, $\Phi_K \in \mathbb{R}^{(N_{\mathrm{tr}}S)\times(3K+1)}$, $Q_K \in \mathbb{R}^{(N_{\mathrm{tr}}S)\times K}$, and $W_K \in \mathbb{R}^{(3K+1)\times K}$.

Target anchor selection. For the target batch, SCG keeps the anchor variant with the largest prototype-bank cosine:

$$(r_K^*, \rho_K^*) = \arg\max_{r,\rho} \max_{p \in P} \cos(a_K^{r,\rho}(x), p), \qquad a_K(x) = a_K^{r_K^*, \rho_K^*}(x). \tag{23}$$

The final readouts use the same candidate family as in the main text, with SCG-Ret selecting from $\{0, 256, 384, 512\}$ and SCG-Recon selecting from $\{0, 64, 128\}$.

C. Intra-Subject SCG Setting and Details

Intra-subject experiments evaluate the backend and calibration family under reduced subject shift.

SCG in the Intra-Subject Setting

When training and test samples come from the same subject, the intra-subject variant uses the same calibration set with statistics from a single subject. The image prototype bank and PCA bases are unchanged. For a training stimulus i, let

$$u_i^K = B_K^\top(x_i - \mu_P), \qquad v_i^K = B_K^\top(p_i - \mu_P) \tag{24}$$

be the EEG coordinates and image prototype coordinates in a K-dimensional PCA subspace of image prototypes. In LOSO, the correction mask is informed by variation across subjects for the same stimulus. In the intra-subject branch, this term is replaced by the coordinate-wise EEG–image distribution gap

$$d^K = \left|\frac{\hat{\mu}_X^K - \hat{\mu}_P^K}{\hat{\sigma}_P^K}\right| + \left|\log \frac{\hat{\sigma}_X^K}{\hat{\sigma}_P^K}\right|, \tag{25}$$

where $\hat{\mu}_X^K, \hat{\sigma}_X^K$ are training EEG coordinate statistics and $\hat{\mu}_P^K, \hat{\sigma}_P^K$ are training statistics of image prototype coordinates. Coordinates with larger d^K are more eligible for calibration, while coordinates with high variance across images or high EEG–image coordinate correlation are protected as semantic axes. The same candidate construction and readout based on prototype confidence are then used at test time with unlabeled test EEG batch statistics.

Intra-Subject Subject-08 Decoding Performance

Figure 7 presents qualitative examples for the Subject-08 intra-subject setting.

Intra-Subject Ablations

Table 7 attributes the intra-subject gain primarily to prior-side OMCR: adding this term increases the semantic metrics more substantially than low-level OMCR, consistent with the direct role of the predicted image embedding in IP-Adapter conditioning. SCG-Recon further improves CLIP while keeping the AlexNet scores and SwAV distance close to the strongest backend values, indicating that prototype-based calibration continues to refine the semantic condition even under reduced subject shift.



Figure 7: Intra-subject Subject-08 reconstruction examples.

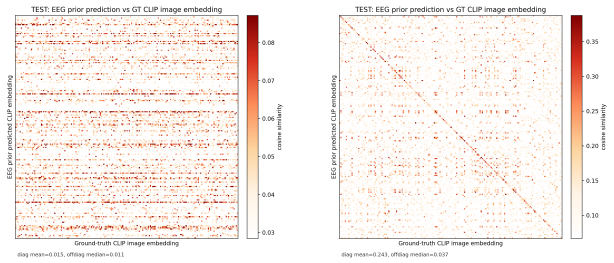
Variant	Alex.2 ↑	Alex.5 ↑	CLIP ↑	SwAV ↓
Base	0.724	0.689	0.576	0.676
+ OMCR(low-level)	0.782	0.740	0.599	0.667
+ OMCR(prior)	0.852	0.899	0.861	0.540
+ OMCR(current)	0.867	**0.916**	0.866	**0.528**
SCG-Recon	**0.868**	0.905	**0.888**	0.538

Table 7: Intra-subject Subject-08 ablation.

Prior term	Alex.2 ↑	Alex.5 ↑	CLIP ↑	SwAV ↓
Cosine only	0.830	0.892	0.825	0.580
Mask contrast only	0.855	0.888	0.843	0.550
Mask contrast + cosine	**0.867**	**0.916**	**0.866**	**0.528**

Table 8: Intra-subject Subject-08 regularizer ablation in the prior branch for OMCR.

Table 8 further decomposes the prior-side objective. Mask contrast improves semantic reconstruction and SwAV relative to cosine-only training, while the cosine anchor stabilizes the paired image target; combining both terms yields the strongest overall prior branch.



Base SCG-Recon

Figure 8: Intra-subject Subject-08 paired similarity.

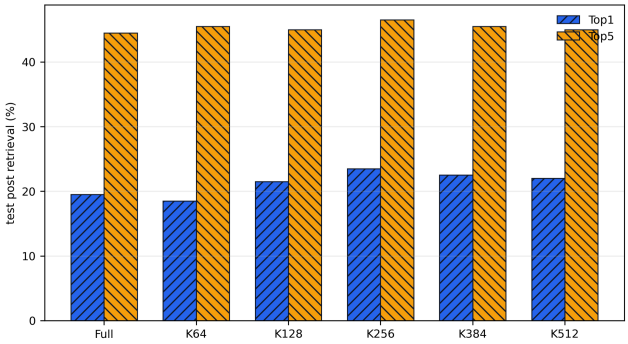


Figure 9: Cross-subject Sub-08 candidate retrieval strength.

Case	Top-1 (%) ↑	Top-5 (%) ↑
Base	15.5	36.0
FullDiag	19.5	44.5
FullCORAL	9.50	30.5
k64	18.5	45.5
k128	21.5	45.0
k256	23.5	**46.5**
k384	22.5	45.5
k512	22.0	45.0
All k-ranks	19.0	45.5
SCG-Ret	**24.0**	46.0

Table 9: Cross-subject Sub-08 SCG retrieval ablation.

Representation Analysis

Figure 8 summarizes the representation change: after SCG-Recon, the paired similarity matrix exhibits a clearer diagonal and weaker off-diagonal background.

D. Cross-Subject SCG Diagnostics

Candidate Readout Trade-off

Figure 9 and Tables 9–10 provide the candidate-level evidence for the readout split: K256 and SCG-Ret favor retrieval, whereas K64/K128 and SCG-Recon favor reconstruction.

Subspace and Selector Diagnostics

These diagnostics examine how the rank split relates to cross-subject semantic structure. Figures 10 and 11 provide additional SCG subspace analyses referenced in the main text.

Case	CLIP ↑	Alex.2 ↑	Alex.5 ↑	SwAV ↓
Base+OMCR	0.679	0.676	0.680	0.631
FullDiag	0.680	0.677	0.696	0.627
FullCORAL	0.696	0.681	0.711	0.632
k64	0.721	0.690	**0.731**	0.617
k128	0.712	0.691	0.727	**0.617**
k256	0.697	**0.692**	0.699	0.620
k384	0.700	0.685	0.720	0.622
k512	0.702	0.692	0.713	0.622
All k-ranks	0.715	0.685	0.729	0.618
SCG-Recon	**0.723**	0.684	0.726	0.618

Table 10: Cross-subject Sub-08 SCG reconstruction ablation.

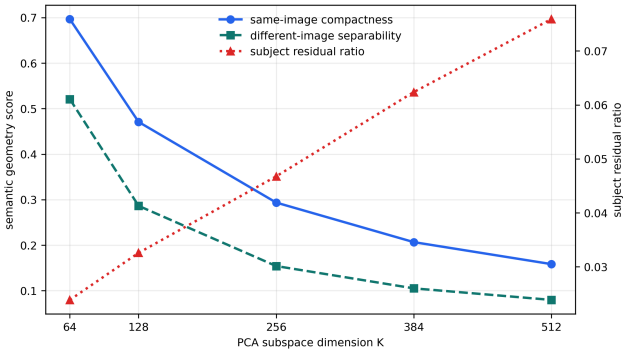


Figure 10: Cross-subject Sub-08 subspace reliability.

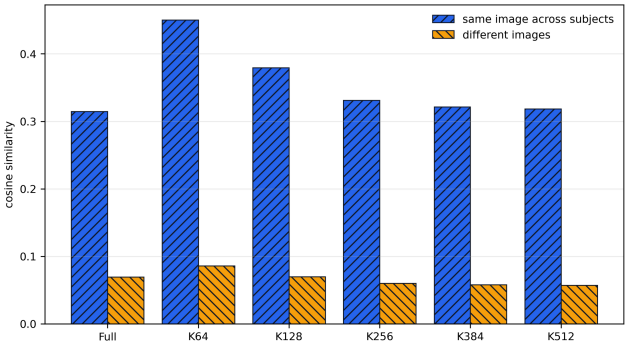


Figure 11: Cross-subject Sub-08 compactness for the same stimulus.

In Figure 10, compactness among responses to the same image and separability between different images are both strongest at low rank, whereas the subject residual ratio increases with the PCA dimension. This pattern indicates that higher-rank coordinates introduce subject-sensitive residual variation more rapidly than stable cross-subject semantic structure. Figure 11 presents the same tendency in pairwise cosine geometry: responses to the same image across subjects remain substantially more similar than responses to different images, with the strongest compactness appearing at the low-rank candidate used by SCG-Recon.

Selector diagnostics evaluate prototype-bank confidence as a sample-wise routing rule for complete semantic candidates. Table 11 reports the low-rank SCG-Recon selector ablation. Fixed uses the best single candidate in the reconstruction readout set, Average uniformly averages eligible candidates, Oracle selects the best candidate by test rank for each sample after evaluation, and Confidence denotes the label-free selector used at inference. Figure 12 visualizes the Confidence computation. For one EEG sample from the target subject, each curve reports the top ten cosine similarities between a candidate semantic vector and the training prototype bank P. The default max-tail10 score is $\mathrm{Conf} = s_{(1)} + 2(s_{(1)} - s_{(10)})$, where $s_{(1)}$ and $s_{(10)}$ are the largest and tenth-largest prototype similarities. This score combines manifold proximity and local sharpness using the training prototype bank.

Readout	Selector	Top-1 (%) ↑	Top-5 (%) ↑
SCG-Recon	Confidence	19.0	45.5
SCG-Recon	Fixed k64	18.5	45.5
SCG-Recon	Average	19.5	45.5
SCG-Recon	Oracle	**27.0**	**49.0**

Table 11: Cross-subject Sub-08 reconstruction selector ablation.

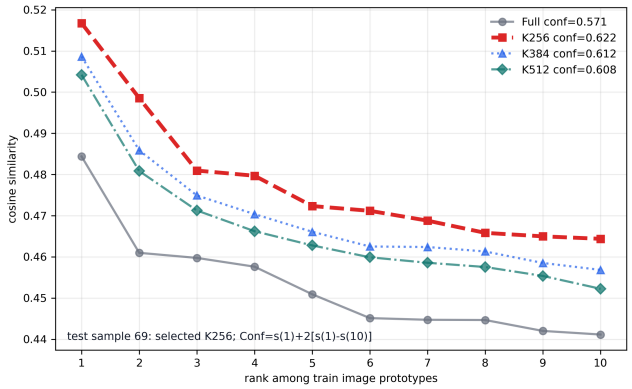


Figure 12: Cross-subject Sub-08 label-free confidence selector.

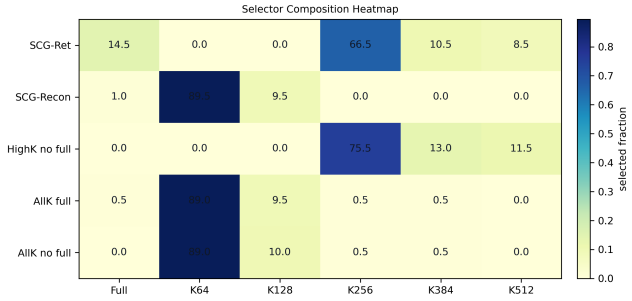


Figure 13: Cross-subject Sub-08 selector composition.

Figure 13 indicates that the confidence selector routes SCG-Ret primarily to the high-rank K256 candidate, while SCG-Recon is routed mainly to K64. Figure 14 indicates that full-dimensional diagonal (Full1024 diag) alignment

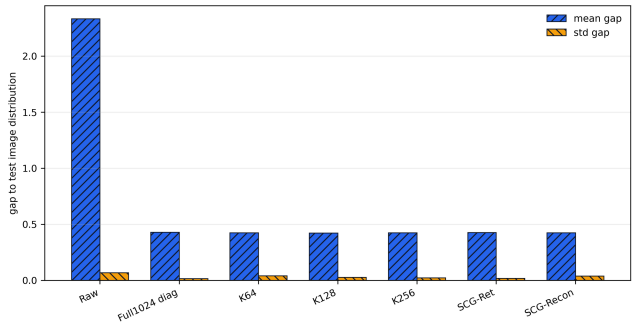


Figure 14: Cross-subject Sub-08 distribution alignment.

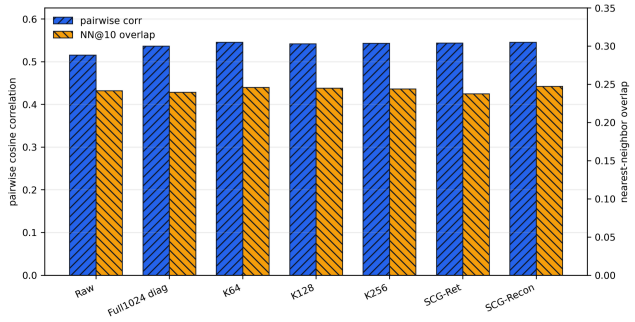


Figure 15: Cross-subject Sub-08 geometry preservation.

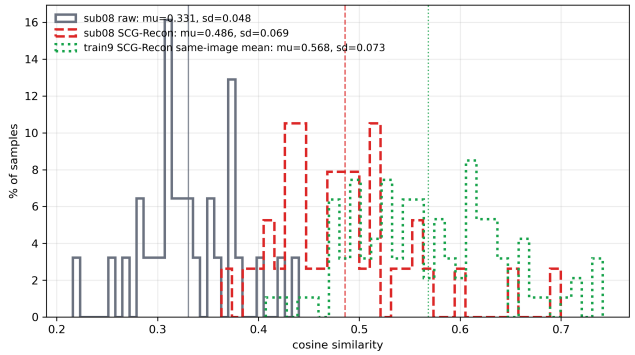


Figure 16: Cross-subject Sub-08 alignment to the image side on Top-1 success samples.

removes most global mean/std mismatch, while Figure 15 shows broadly preserved pairwise geometry with limited local nearest-neighbor overlap. Figure 16 repeats the cosine analysis against the paired image on samples with successful Top-1 retrieval. The oracle rows in Tables 11 quantify the remaining selection gap, whereas the confidence rows correspond to the label-free readouts used at inference.

Prior Replacement Diagnostic

The replacement study probes reusable cross-subject image-specific structure through controlled source-subject substitution. Table 12 reports the numeric prior-replacement results corresponding to Figure 17. Nearest-neighbor replacement provides a modest reconstruction gain at a moderate replacement ratio, whereas full nearest-neighbor replacement reduces Top-1 retrieval. Oracle replacement improves all semantic reconstruction metrics by using source-subject responses to the same image, thereby estimating the amount of image-specific structure available across subjects. Figure 17 plots the same intervention as retrieval–reconstruction points: blue points are NN replacement ratios, red points are oracle means over responses to the same image, the gray square is Raw, and the black diamond is SCG-Recon. The plot places SCG-Recon near the best NN reconstruction point while retaining higher retrieval than high-ratio NN replacement. Together, the table and plot indicate that direct source substitution and target-subject calibration capture different cross-subject signals, with the oracle row serving as an analysis reference for shared image-specific information.

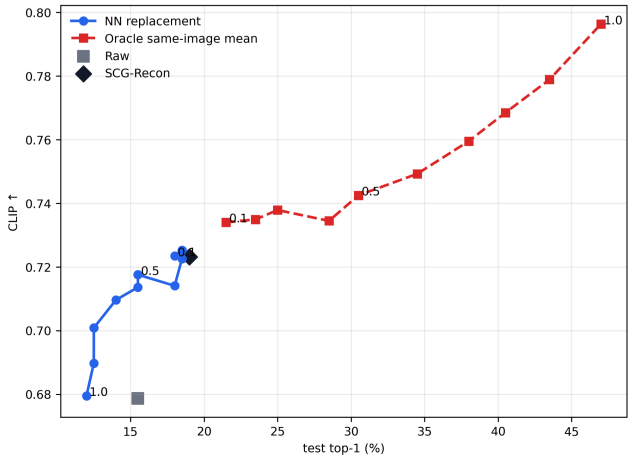


Figure 17: Cross-subject Sub-08 prior replacement trade-off.

Method	Ratio	Top-1 ↑	CLIP ↑	Alex.2 ↑	Alex.5 ↑	SwAV ↓
Base+OMCR	–	15.5	0.679	0.676	0.680	0.631
SCG-Recon	–	19.0	0.723	0.684	0.726	0.618
NN replace	0.200	18.5	0.725	0.689	0.734	0.612
NN replace	1.00	12.0	0.679	0.697	0.723	0.625
Oracle	0.500	30.5	0.743	0.712	0.756	0.595
Oracle	1.00	**47.0**	**0.796**	**0.765**	**0.828**	**0.564**

Table 12: Cross-subject Sub-08 prior replacement analysis.

β	Top-1 (%) ↑	Top-5 (%) ↑
0.0625	22.5	45.5
0.125	**24.0**	46.0
0.250	22.5	**46.5**
0.500	20.0	45.5

Table 13: Branch blend (β) sensitivity.

Implementation Sensitivity Checks

Tables 13 and 14 record SCG sensitivity runs within the same Sub-08 setting. The results show stable behavior around the reported configuration and serve as implementation diagnos-

α	CLIP ↑	Alex.2 ↑	Alex.5 ↑	SwAV ↓
SCG-Ret:0.10	**0.694**	0.678	0.702	0.622
0.00	0.690	0.676	0.699	**0.620**
0.05	0.693	0.674	**0.708**	0.621
0.20	0.688	**0.679**	0.700	0.622

Table 14: Image teacher interpolation (α) sensitivity.

Subject	PixCorr ↑	SSIM ↑	Alex.2 ↑	Alex.5 ↑	CLIP ↑	SwAV ↓
Sub-01	0.152	0.350	0.678	0.729	0.703	0.628
Sub-02	0.149	0.367	**0.752**	0.763	0.710	0.607
Sub-03	0.128	0.374	0.730	0.750	0.732	0.614
Sub-04	0.138	0.348	0.670	0.719	0.693	0.630
Sub-05	0.159	0.366	0.720	0.753	0.696	0.623
Sub-06	0.154	0.368	0.678	0.730	0.704	0.612
Sub-07	0.134	0.347	0.682	0.736	0.703	0.632
Sub-08	**0.174**	**0.389**	0.684	0.726	0.723	0.618
Sub-09	0.156	0.381	0.723	0.744	0.682	0.618
Sub-10	0.147	0.359	0.737	**0.803**	**0.781**	**0.601**
Mean	0.149	0.365	0.706	0.745	0.713	0.618

Table 15: Cross-subject SCG-Recon under ten-subject LOSO.

Method	Stat	Alex.2 ↑	Alex.5 ↑	CLIP ↑	SwAV ↓
Base+OMCR	Mean	0.681	0.701	0.656	0.640
	Std	0.012	0.023	0.030	0.016
FullDiag	Mean	0.699	0.731	0.695	0.623
	Std	0.023	0.024	0.035	0.014
SCG-Ret	Mean	0.695	0.724	0.688	0.625
	Std	0.023	0.031	0.038	0.012
SCG-Recon	Mean	**0.706**	**0.745**	**0.713**	**0.618**
	Std	0.028	0.023	0.027	0.010

Table 16: Ten-subject LOSO cross-subject reconstruction summary, reported as mean and population standard deviation.

tics for the candidate construction.

E. Intra-Subject and Cross-Subject Reconstruction Metrics

These tables provide the complete reconstruction results associated with the main decoding analysis. Tables 15 and 16 report cross-subject LOSO reconstruction by subject and by SCG readout, while Table 17 reports the corresponding intra-subject SCG-Recon results.

F. Intra-Subject and Cross-Subject Retrieval Accuracy

Compact retrieval summary. Retrieval provides an additional measure of semantic alignment. Table 18 reports the ten-subject mean and population standard deviation, and Table 19 provides the intra-subject breakdown by subject. NeuroBridge (ViT-H-14) is evaluated in the same semantic space used by our reconstruction pipeline.

Subject	PixCorr ↑	SSIM ↑	Alex.2 ↑	Alex.5 ↑	CLIP ↑	SwAV ↓
Sub-01	0.252	0.382	0.849	0.879	0.812	0.583
Sub-02	0.242	0.384	0.829	0.884	0.835	0.564
Sub-03	0.249	0.375	0.845	0.872	0.858	0.570
Sub-04	0.228	0.371	0.832	0.880	0.861	0.556
Sub-05	0.223	0.368	0.834	0.858	0.809	0.579
Sub-06	0.261	0.376	0.870	0.901	0.858	0.555
Sub-07	0.230	0.372	0.819	0.862	0.859	0.561
Sub-08	**0.304**	**0.397**	0.868	0.905	**0.888**	0.538
Sub-09	0.264	0.387	0.836	0.875	0.827	0.560
Sub-10	0.262	0.382	**0.871**	**0.920**	0.886	**0.531**
Mean	0.251	0.380	0.845	0.883	0.849	0.560

Table 17: Intra-subject SCG-Recon.

Method	Intra-subject		Cross-subject	
	Top-1 (%) ↑	Top-5 (%) ↑	Top-1 (%) ↑	Top-5 (%) ↑
NICE	16.1 ± 3.51	43.6 ± 6.52	6.20 ± 1.06	21.4 ± 2.82
ATM	27.1 ± 7.02	58.1 ± 7.74	11.9 ± 4.56	33.8 ± 8.27
AVDE	30.0 ± 5.90	58.2 ± 7.60	14.3 ± 3.57	32.9 ± 6.61
NeuroBridge (ViT-H-14)	50.9 ± 6.14	82.4 ± 6.73	16.4 ± 4.20	39.5 ± 6.64
FullDiag	51.2 ± 6.24	**83.9** ± 5.55	22.5 ± 3.88	**50.1** ± 5.72
SCG-Ret	**52.8** ± 6.49	83.6 ± 5.91	**23.7** ± 3.80	49.2 ± 5.72
SCG-Recon	51.9 ± 5.97	82.8 ± 6.04	23.3 ± 3.87	49.5 ± 4.46

Table 18: Mean intra-subject and cross-subject retrieval accuracy, reported as mean ± population standard deviation.

In the cross-subject setting, SCG-Ret achieves the highest mean Top-1 accuracy, while the diagonal candidate gives the highest Top-5 accuracy.

G. Additional Qualitative Reconstructions

Limitations and future work. Figure 18 extends the qualitative comparison across food, plants, animals, objects, clothing, and transportation, whereas Figure 19 presents challenging samples selected from the remaining reconstructions. Errors range from complete category loss to preservation of a coarse cue, such as clothing or transportation, without the object identity. Furniture and electronic devices further expose weak recovery of target-specific appearance and geometry. The visual coherence of these outputs, despite their semantic mismatch, shows that image realism alone does not ensure semantic fidelity when fine-grained EEG evidence from a held-out subject is ambiguous.

Future work may refine both the neural representation and its interface with generation. Modeling EEG frequency and temporal dynamics with electrode topology, together with reliability-weighted trial aggregation, could stabilize the inputs to SCG. Candidate weighting informed by uncertainty and multiscale coordinates could improve semantic precision, while spatial guidance in the low-level branch could preserve object geometry. Additional subjects and recording sessions would assess these extensions under zero-shot transfer.

Method	Metric (%)	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10	Mean
Intra-subject: train and test on one subject												
NICE	Top-1	13.2	13.5	14.5	20.6	10.1	16.5	17.0	22.9	15.4	17.4	16.1
	Top-5	39.5	40.3	42.7	52.7	31.5	44.0	42.1	56.1	41.6	45.8	43.6
ATM	Top-1	25.6	22.0	25.0	31.4	12.9	21.3	30.5	38.8	34.4	29.1	27.1
	Top-5	60.4	54.5	62.4	60.9	43.0	51.1	61.5	72.0	51.5	63.5	58.1
AVDE	Top-1	25.0	24.1	27.5	29.8	25.4	33.5	27.4	41.7	26.1	39.5	30.0
	Top-5	55.2	51.0	58.6	54.7	50.3	60.3	55.2	71.3	52.1	73.0	58.2
NeuroBridge (ViT-H-14)	Top-1	43.5	**53.5**	51.0	52.0	41.0	56.0	47.5	58.5	45.5	**60.5**	50.9
	Top-5	71.5	79.0	**86.0**	**83.5**	70.5	87.5	83.5	88.5	**81.5**	**92.5**	82.4
SCG-Ret (Ours)	Top-1	**48.5**	52.0	**52.5**	**53.5**	**41.5**	**60.5**	**49.0**	**63.5**	**47.0**	60.0	**52.8**
	Top-5	**77.5**	**81.0**	85.5	**83.5**	**71.0**	**89.5**	**86.5**	**89.5**	80.5	91.0	**83.6**

Table 19: Complete intra-subject 200-way retrieval accuracy.

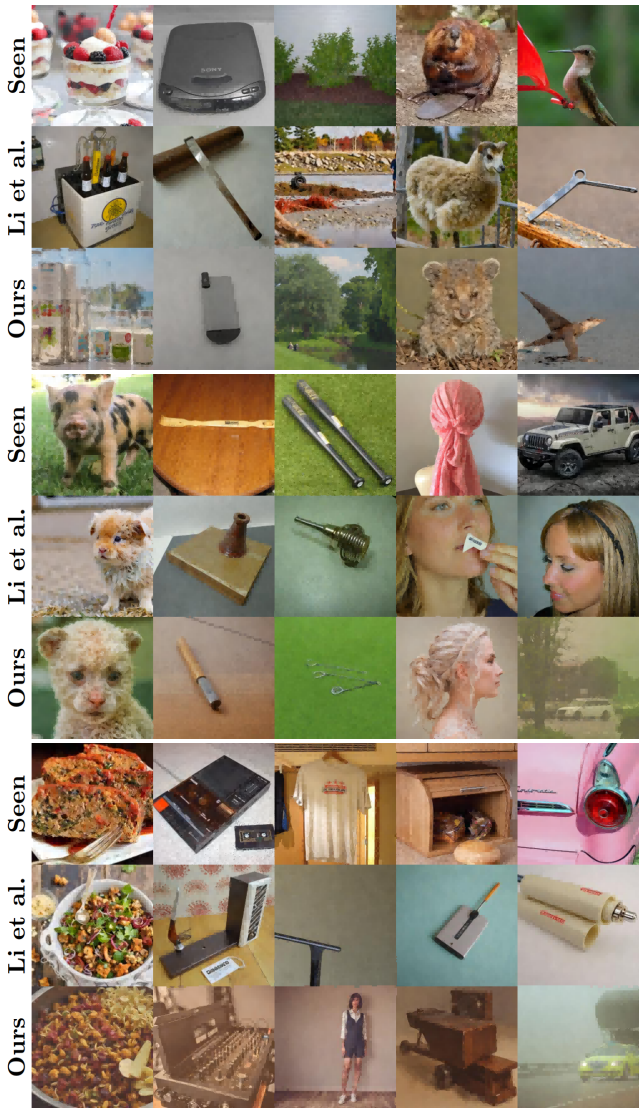


Figure 18: Additional cross-subject Sub-08 qualitative reconstruction examples.

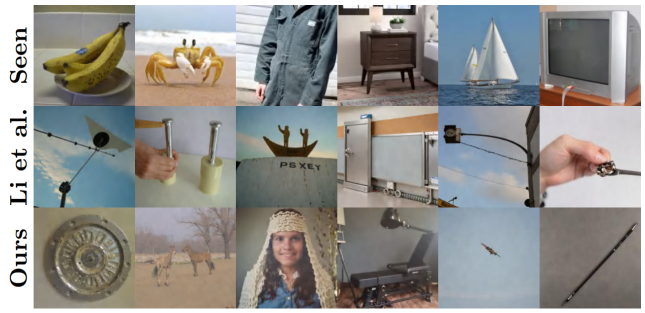


Figure 19: Representative difficult cases from the remaining cross-subject Sub-08 reconstructions. Rows compare the seen stimulus, Li et al., and our reconstruction.